

# Project 4: Preference Elicitation

Design and Implementation Report

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This report documents the feature representation and preference model used by the interactive interface (Tasks 1 and 2), and the design of a user study comparing two elicitation interfaces (Task 3). The interactive elicitation interface itself (Task 4) is available from the project landing page.

## Task 1 — Feature Representation

Movie features combine a quality signal (IMDB score), scale (log budget, log gross revenue, log vote count), temporal context (release year), popularity (log Facebook likes), and genre indicators for the eight most common genres. These attributes are interpretable, available for nearly all movies in the dataset, and align with factors users commonly cite when explaining their movie preferences. Monetary and count variables are log-transformed to reduce the influence of extreme outliers (e.g. blockbuster budgets), and all features are standardized to zero mean and unit variance so that the fitted weight vector  $w$  is comparable across features.

### Feature vector components

- IMDB score (quality signal)
- Duration in minutes
- Log budget
- Log gross revenue
- Log number of votes
- Release year (normalized)
- Movie Facebook likes (log)
- Genre indicators for major categories

**Genre indicators** (one binary feature per genre): Action, Adventure, Comedy, Drama, Romance, Thriller, Crime, Sci-Fi.

Each movie is represented by a standardized numeric feature vector  $x$ . User utility is modeled as a linear function of these features:  $U(x) = w^T x$ , where  $w$  is the user's latent preference vector estimated from their responses.

## Task 2 — Ranking Extension of Bradley-Terry

We extend the Bradley-Terry model to full rankings using the Plackett-Luce model. For a ranking  $i_1 > i_2 > \dots > i_n$  of  $n$  items (read ">" as "preferred to"), we decompose the probability of observing the entire order as a sequence of independent choices, each picking the most preferred item among those not yet ranked:

$P(\text{ranking}) = [\text{product over } k = 1 \dots n \text{ of}] \exp(u_{i_k})$  divided by the sum of  $\exp(u_{i_j})$  over all items  $i_j$  still remaining at step  $k$ ,

where  $u_i = w^T x_i$  is the latent utility of item  $i$ . Each factor in the product is itself a standard Bradley-Terry choice probability among the items not yet ranked, so pairwise comparison (Design 1) is exactly the special case  $n = 2$  (a single choice between two remaining items). This formulation lets both interfaces estimate the same latent vector  $w$  with the same likelihood family, which is what makes a direct comparison between them meaningful.

In this project, Design 1 (pairwise comparisons) uses the standard two-item Bradley-Terry likelihood, and Design 2 (ranking of ten movies) uses the Plackett-Luce likelihood for full orderings described above. Both interfaces ultimately estimate the same latent vector  $w$ , which is what makes the two designs directly comparable.

## Parameter estimation

The preference vector  $w$  is estimated by maximizing the Bradley-Terry / Plackett-Luce log-likelihood of the observed choices, plus an L2 (ridge) penalty on  $w$ . With as few as 15 pairwise comparisons and roughly 15 features, the unregularized maximum-likelihood estimate is only weakly identified: if a participant's choices happen to be perfectly separable along some direction in feature space, the unregularized estimate diverges to infinite magnitude, a well-known degeneracy of logistic-type likelihoods on separable data. The L2 penalty corresponds to a Gaussian prior on  $w$  and keeps the estimate finite and numerically stable even after a small number of responses.

## Task 3 — User Study Protocol

### Comparing Pairwise and Ranking Preference Elicitation for Movie Recommendation

H1: The ranking interface (Design 2) yields more accurate estimates of the latent preference vector  $w$  than the pairwise interface (Design 1), at the cost of longer interaction time and higher cognitive load.

#### Study overview

|                               |                                                                               |
|-------------------------------|-------------------------------------------------------------------------------|
| <b>Design type</b>            | Between-subjects randomized controlled study                                  |
| <b>Design 1 (condition A)</b> | 15 pairwise comparisons of two movies each                                    |
| <b>Design 2 (condition B)</b> | 2 ranking tasks with 10 movies each                                           |
| <b>Target sample size</b>     | 40                                                                            |
| <b>Inclusion criteria</b>     | Adults who watch movies at least occasionally and can read English.           |
| <b>Recruitment</b>            | University mailing lists, social media, or Prolific with a fair compensation. |

#### Procedure

- 1 Obtain informed consent and collect demographic information.
- 2 Randomly assign participant to Design 1 or Design 2.
- 3 Show standardized instructions with one practice trial.
- 4 Run the elicitation tasks with movies sampled uniformly at random.
- 5 Collect task completion time and optional NASA-TLX workload ratings.
- 6 Estimate  $w$  with the Bradley-Terry / Plackett-Luce model.
- 7 Evaluate on holdout pairwise comparisons not shown during elicitation.
- 8 Debrief participants and provide optional recommendation preview.

#### Outcome metrics

- Holdout pairwise prediction accuracy (Bradley-Terry log-likelihood on unseen pairs).
- Top-k recommendation overlap with participant validation choices.
- Total study duration and time per task.
- Subjective workload and preference confidence (post-study questionnaire).

#### Analysis plan

- Compare holdout accuracy between Design 1 and Design 2 using a two-sample t-test.
- Compare completion time with Mann-Whitney U test.
- Report effect sizes and confidence intervals.
- Perform sanity checks for random responding and incomplete sessions.

#### Known limitations and confounds

- The IMDB score is deliberately hidden from participants during the elicitation tasks, even though it is used as a model feature. Showing a crowd rating while asking for personal preference risks anchoring participants on the aggregate opinion of others rather than eliciting their own taste.
- Movies are sampled uniformly at random rather than adaptively; this keeps the comparison between designs fair but is not the most sample-efficient elicitation strategy (see the note on adaptive selection in the task description).
- Order effects (fatigue, learning) are not counterbalanced across rounds in this demonstration interface; a full deployment should randomize or counterbalance round order.